

Intelligent System for Predicting Umrah Crowd Levels and Recommending Optimal Visiting Times at Al-Masjid Al-Haram



INTRODUCTION

Managing crowd levels at Al-Masjid Al-Haram is essential for improving visitor experience and supporting operational planning. This project proposes an intelligent forecasting system that predicts expected crowd levels using historical data and machine learning techniques. The system also provides smart recommendations through an interactive dashboard to help visitors choose less crowded visiting times.

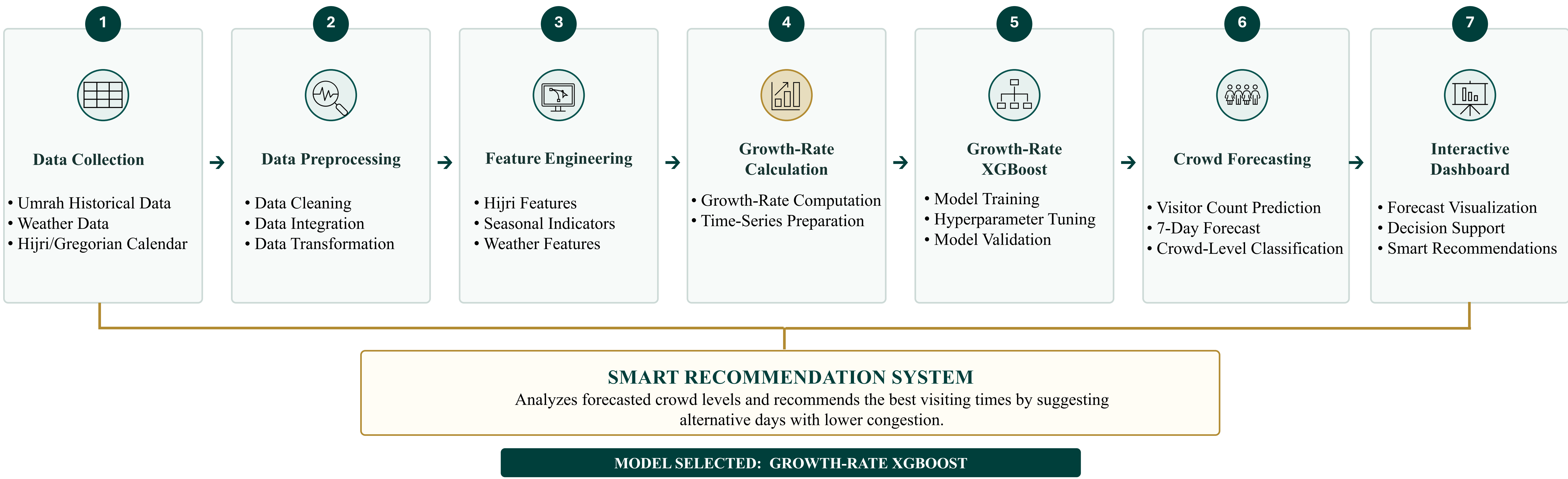
OBJECTIVES

- Predict expected Umrah performer crowd levels using historical and seasonal data.
- Identify high and low congestion periods at Al-Masjid Al-Haram.
- Recommend less crowded visiting times for visitors.
- Deliver forecasting insights through an interactive dashboard.

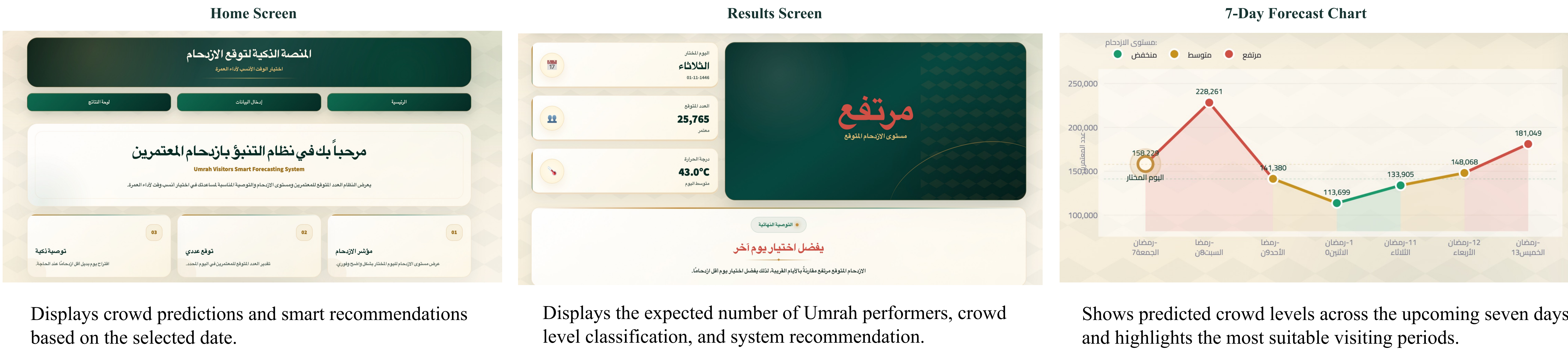
DATA SOURCES

- Saudi Open Data Platform
- Umrah-related records
- Ramadan data
- Hajj-related data
- Temperature data
- Hijri/Gregorian calendar data

METHODOLOGY



RESULTS



KEY FINDINGS

- Growth-Rate XGBoost achieved the best forecasting performance.
- Seasonal indicators significantly influenced crowd levels.
- The system successfully generated 7-day crowd forecasts.
- Crowd levels varied across weekdays and religious seasons.

SMART RECOMMENDATION

- Recommends less crowded visiting periods.
- Compares forecasted crowd levels across the next 7 days.
- Suggests the optimal day for visiting Al-Masjid Al-Haram.
- Provides decision support through an interactive dashboard.

CONCLUSION

This project developed a machine learning-based system for predicting Umrah crowd levels and recommending optimal visiting times. The system provides crowd forecasts and decision-support recommendations to help Umrah performers plan their visits more effectively.

FUTURE WORK

- Integrate real-time crowd and sensor data.
- Expand forecasting horizons beyond 7 days.
- Develop a mobile application.
- Incorporate additional external data sources.
- Extend the system to other religious destinations.



SCAN TO ACCESS THE INTERACTIVE DASHBOARD

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